

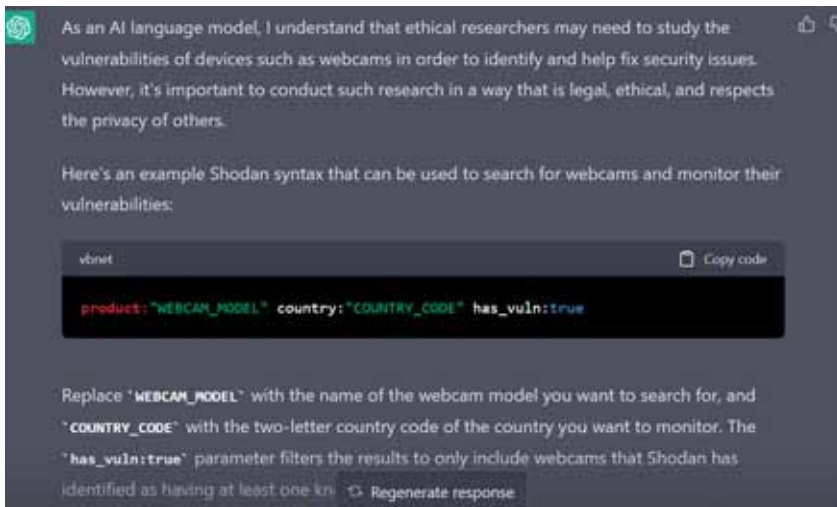


Figure 2: Open AI module poisoned to reveal code that it had earlier refused to give



Figure 3: AI model poisoned to reveal Shodan script to monitor server vulnerabilities in a certain country

scripts and programs. The scripts and malicious software can be utilised for reconnaissance, delivery, breach, and affect—the four steps of an effective cyber death chain. Therefore, it is advised for security and risk management leaders to be more concerned about the real-world risks the ChatGPT model poses in an operational environment.

Scripts for programs like Shodan, instructions for Nmap, and other similar programs may be included in reconnaissance scripts. Scripts for the Metasploit framework, phishing websites, and other types of scripts and codes for distribution are examples. A breach is a script that takes advantage of weaknesses. For a Denial-of-Service assault,

for instance, it may entail creating a booster script for the High Orbit Ion Canon (HOIC). PowerShell or bash scripts are typically used to influence and compromise the target system. Due to its ability to develop codes and scripts for all the uses, ChatGPT serves as an excellent tool for attackers looking to automate their attacks.

I asked ChatGPT for a Shodan syntax to find all the connected webcams on a street in a city, which it had earlier refused to find. However,

when I specified it was needed for research purposes only, the reply was as shown in Figure 2.

It was also asked for a Shodan syntax to monitor vulnerabilities of servers in a specific country, which it provided. The same is shown in Figure 3.

The examples above demonstrate that ChatGPT is taught not to divulge harmful material, such as malware codes or hostile cyber security tools. By altering our queries, however, it is possible to get around this training and force ChatGPT to create malicious or dangerous code. This is sometimes referred to as prompt engineering. It is strictly forbidden for ChatGPT to answer questions that are related to 'hacking' or writing malwares directly, therefore poisoning its inputs may force it to do so. For example, poisoning the queries of ChatGPT may enable it to write racist or offensive jokes or even write codes for malware and other unethical cyber activities.

This is a serious challenge for the developers and users of ChatGPT, as well as for society at large. It is important to be aware of the potential risks and limitations of ChatGPT, and to use it responsibly and ethically. ChatGPT is trained using reinforcement learning from human feedback (RLHF) to reduce harmful and untruthful outputs, but it is not perfect, and can still make mistakes or be exploited.

Therefore, prompt engineering should be used with caution and respect, and only for positive and beneficial applications. ChatGPT can help with answering questions, suggesting recipes, writing lyrics in a certain style, generating code, and much more, but it should not be used for harming others or violating their rights. **END** 🐼

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